
Session 2 (2h) — Examples, Roles & Output Formatting

Core concepts (1h)

- **Zero-shot vs few-shot prompting:**
 - Zero-shot: just ask, no examples (“Summarize this in 2 sentences”)
 - Few-shot: show 1–3 examples of input → desired output first, then give the real input. Massively improves consistency for repetitive tasks (e.g., “convert 10 sensor readings into this exact CSV format”).
- **Role/persona framing** — telling the AI *who* to act as (“act as a senior firmware reviewer”) shifts the vocabulary and depth of the answer, but doesn’t add facts it doesn’t have — it’s a formatting/framing tool, not a knowledge upgrade.
- **Requesting explicit output format** — asking for a specific structure (a table, numbered steps, a code block, a fixed set of headings) makes responses far easier to drop directly into a report or a teammate’s document, instead of having to reformat manually every time.
- **Negative constraints** — telling the AI what to avoid (“don’t include background theory, just the fix”) is often more effective than only describing what you want.

Hands-on practice (1h)

1. Take 3 raw sensor readings or measurements from any of your other prep packs.
2. Write a few-shot prompt: show 1 example of “raw value → formatted table row”, then ask it to format the other readings the same way.
3. Ask the same underlying question twice — once with no role framing, once with “act as a hardware reviewer checking my documentation for an internship submission” — compare how the tone/focus shifts.
4. Practice one explicit-format request: “give me this as a 5-row markdown table with columns Symptom / Likely Cause / Fix” for a real troubleshooting question.

Checkpoint

- ☐ Can explain zero-shot vs few-shot and when few-shot is worth the extra setup
- ☐ Used role framing and can describe what it changed (and what it didn’t)
- ☐ Successfully requested a specific output format (table/list/code block) and got it